

Thin-Sheet Error and the TAG Mass Budget

Giovanni Fereoli 6 October 2026

Equation labels in sans serif (e.g. **z0**) refer to the dissertation [D]. z points up, $k = k_{mn} = j_{mn}/(\alpha R^*)$ is the wavenumber of mode (m, n) and $\lambda = 2\pi/k$ its wavelength.

Why the error is larger than σ

The mass is linear in the data: the fit returns $\Delta\hat{\mathbf{c}} = A^+ \Delta\mathbf{d}$ (A^+ the truncated-SVD inverse) and Wahr's inversion sums it, $\Delta\hat{M} = \mathbf{f}_M^T \Delta\hat{\mathbf{c}}$. Split the data into the field the model assumes ($\mathbf{d}_0$: the true $\Delta\varrho$ as a flat sheet on z_0), what the assumptions leave out ($\delta\mathbf{d}$), and noise ($\mathbf{n}$). Then

$$\Delta\hat{M} - \Delta M = \underbrace{\mathbf{f}_M^T A^+ \mathbf{d}_0 - \Delta M}_{b_{\text{basis}}} + \underbrace{\mathbf{f}_M^T A^+ \delta\mathbf{d}}_{b_{\text{wa}} + b_{\text{th}} + \dots} + \underbrace{\mathbf{f}_M^T A^+ \mathbf{n}}_{\text{noise}}, \quad \sigma^2 = \mathbf{f}_M^T A^+ \Sigma_n A^+ \mathbf{f}_M. \quad (1)$$

The formal σ contains only the noise. The b_i are biases: they are the same every time, so no covariance sees them. The triangle inequality bounds the error by

$$|\Delta\hat{M} - \Delta M| \leq \sigma_{\text{sys}} + 2\sigma, \quad \sigma_{\text{sys}} = \sum_i |b_i|, \quad \text{report } \Delta\hat{M} \pm \sigma \text{ (stat)} \pm \sigma_{\text{sys}} \text{ (sys)}. \quad (2)$$

In words: σ says how far the answer would move if the noise were drawn again. A wrong assumption moves it the same way every time, so σ cannot see it. Each assumption adds a bias b_i : the estimator applied to the signal that the assumption leaves out. The Result below gives that signal for the two thin-sheet assumptions, wander and thickness, mode by mode ($\delta\mathcal{C}^i = \varepsilon_k \mathcal{C}^i$). The mass-kernel section turns every b_i into one integral and bounds it; the TAG section gives the numbers.

Result

A column of moved material with thickness Δh and middle at height ζ above the sheet plane z_0 is returned by the thin-sheet model with the relative error

$$\varepsilon_k = e^{k\zeta} S(k\Delta h) - 1 = \underbrace{k\zeta + \frac{1}{2}(k\zeta)^2}_{\text{wander}} + \underbrace{\frac{1}{24}(k\Delta h)^2 + O(k^3)}_{\text{thickness}}, \quad S(u) = \frac{\sinh(u/2)}{u/2}. \quad (3)$$

The two parts alone are

$$\varepsilon_k^{\text{wa}} = e^{k\zeta} - 1 = k\zeta + \frac{(k\zeta)^2}{2} + \frac{(k\zeta)^3}{6} + \dots = \frac{2\pi\zeta}{\lambda} + \dots \quad (4)$$

$$\varepsilon_k^{\text{th}} = S(k\Delta h) - 1 = \frac{(k\Delta h)^2}{24} + \frac{(k\Delta h)^4}{1920} + \dots = \frac{1}{24} \left(\frac{2\pi\Delta h}{\lambda} \right)^2 + \dots \quad (5)$$

For the total moved mass the wander cancels at first order, but only for modes much longer than the footprint:

$$\varepsilon_M = \frac{1}{2}(ks)^2 + O(k^3) \quad \text{if } kR^* \ll 1, \quad s^2 = \frac{\int \Delta\varrho (\zeta^2 + \Delta h^2/12) dA}{\int \Delta\varrho dA}. \quad (6)$$

Proof

Step 1: the model. The moved mass is a sheet of surface density $\Delta\varrho$ on $z = z_0$, and each coefficient becomes a modal density by Eq. `sigma_amn`, $\hat{\sigma}_{mn}^c = k \mathcal{C}_{mn}^i / (2\pi G)$.

Step 2: one lifted sheet. Mode (m, n) of a sheet on z_0 fades upward as $\mathcal{U}_{mn} \propto e^{-k(z-z_0)}$. The same sheet at $z_0 + \zeta'$ gives

$$e^{-k(z-z_0-\zeta')} = e^{k\zeta'} e^{-k(z-z_0)}. \quad (7)$$

The height dependence $e^{-k(z-z_0)}$ is the one the model expects, so the fit returns the coefficient of a sheet on z_0 that is $e^{k\zeta'}$ times heavier: $\hat{\sigma} = e^{k\zeta'} \sigma$. The field height z cancels. *In words:* each mode is a wave that fades upward, and short waves fade fastest. Mass placed higher is closer to the field points and looks heavier.

Step 3: a column is a stack of sheets. The moved material in a column has uniform density between h_{pre} and h_{post} (Eq. `sigma_h`): thickness $\Delta h = h_{\text{post}} - h_{\text{pre}}$, middle at $\zeta = \frac{1}{2}(h_{\text{pre}} + h_{\text{post}}) - z_0$ (Eq. `zeta`). Each slice $d\zeta'$ carries the fraction $d\zeta'/\Delta h$ of the column and is returned with the gain of Step 2. Adding the slices,

$$G_k = \frac{1}{\Delta h} \int_{\zeta-\Delta h/2}^{\zeta+\Delta h/2} e^{k\zeta'} d\zeta' = \frac{e^{k(\zeta+\Delta h/2)} - e^{k(\zeta-\Delta h/2)}}{k \Delta h} = e^{k\zeta} \frac{e^{k\Delta h/2} - e^{-k\Delta h/2}}{k \Delta h} = e^{k\zeta} S(k\Delta h), \quad (8)$$

using $e^y - e^{-y} = 2 \sinh y$ with $y = k\Delta h/2$. This is exact, and $\varepsilon_k = G_k - 1$ proves the first equality of (3). *In words:* the gain of the column is the average gain of its slices: one factor for where its middle is (wander), one for how thick it is (thickness).

Step 4: expand. With $x = k\zeta$ and $u = k\Delta h$,

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots \quad (9)$$

$$S(u) = \frac{\frac{u}{2} + \frac{1}{3!}\left(\frac{u}{2}\right)^3 + \frac{1}{5!}\left(\frac{u}{2}\right)^5 + \dots}{u/2} = 1 + \frac{1}{6}\left(\frac{u}{2}\right)^2 + \frac{1}{120}\left(\frac{u}{2}\right)^4 + \dots = 1 + \frac{u^2}{24} + \frac{u^4}{1920} + \dots \quad (10)$$

The second line is (5). Multiplying the two,

$$G_k = \left(1 + k\zeta + \frac{1}{2}k^2\zeta^2 + \dots\right) \left(1 + \frac{1}{24}k^2\Delta h^2 + \dots\right) = 1 + k\zeta + \frac{1}{2}k^2\zeta^2 + \frac{1}{24}k^2\Delta h^2 + O(k^3), \quad (11)$$

which is (3). The $O(k^3)$ remainder is $\frac{1}{6}(k\zeta)^3 + \frac{1}{24}(k\zeta)(k\Delta h)^2$. $\square$

Why 1/24. Expand the gain inside the integral instead, $e^{k\zeta'} = 1 + k\zeta' + \frac{1}{2}k^2\zeta'^2 + \dots$, and average over the column, which is uniform on $[\zeta - \Delta h/2, \zeta + \Delta h/2]$:

$$\langle \zeta' \rangle = \zeta, \quad \langle \zeta'^2 \rangle = \frac{(\zeta + \Delta h/2)^3 - (\zeta - \Delta h/2)^3}{3 \Delta h} = \zeta^2 + \frac{\Delta h^2}{12}, \quad (12)$$

$$G_k = 1 + k\zeta + \frac{1}{2}k^2\left(\zeta^2 + \frac{\Delta h^2}{12}\right) + \dots, \quad \frac{1}{2} \cdot \frac{1}{12} = \frac{1}{24}. \quad (13)$$

In words: $\Delta h^2/12$ is the variance of a uniform layer. The top half of the column looks too heavy and the bottom half too light; the first-order parts cancel, and only the always-positive second-order part survives. So thickness is a small, second-order error that always over-estimates. The middle height ζ enters at first order: that is the wander, and it is why the wander condition below is much stricter.

Step 5: total mass. ΔM is carried by the zonal modes $(0, n)$, and the error of each is $\int \Delta \varrho (G_k - 1) J_0(k\rho) dA$. With $G_k - 1 = k\zeta + \frac{1}{2}k^2(\zeta^2 + \Delta h^2/12) + O(k^3)$ and $J_0(x) = 1 - \frac{1}{4}x^2 + \dots$,

$$\int \Delta \varrho (G_k - 1) J_0 dA = k \int \Delta \varrho \zeta dA + \frac{1}{2}k^2 \int \Delta \varrho \left(\zeta^2 + \frac{\Delta h^2}{12}\right) dA - \frac{1}{4}k^3 \int \Delta \varrho \zeta \rho^2 dA + \dots \quad (14)$$

Eq. `z0` puts z_0 at the centroid of the signed slab: since $\frac{1}{2}(h_{\text{post}}^2 - h_{\text{pre}}^2) = \Delta h (\zeta + z_0)$, it reads $\sum \Delta h (\zeta + z_0) = z_0 \sum \Delta h$, i.e. $\sum \Delta h \zeta = 0$. So $\int \Delta \varrho \zeta dA = 0$ and the first term vanishes. Dividing by $\int \Delta \varrho dA$ gives (6). The third term is the wander again, times $(k\rho)^2$: it is small only when $kR^* \ll 1$. $\square$ *In words:* the sheet sits at the centre of mass, so for the total mass the too-heavy high material and the too-light low material cancel. This holds only for a mode that is flat across the footprint; a shorter mode weights the centre and the rim differently, and the cancellation is lost.

When the thin sheet is fine

Ask each error in (3) to stay below a tolerance ε . Both grow with k , so only the shortest mode needs checking, $k_{\max} = k_{M-1,N} = 2\pi/\lambda_{\min}$. The leading terms of (4) and (5) give

$$\boxed{\underbrace{|\zeta| \leq \frac{\varepsilon}{k_{\max}} = \frac{\varepsilon}{2\pi} \lambda_{\min}}_{\text{wander}}, \quad \underbrace{|\Delta h| \leq \frac{\sqrt{24\varepsilon}}{k_{\max}} = \frac{\sqrt{24\varepsilon}}{2\pi} \lambda_{\min}}_{\text{thickness}}.} \quad (15)$$

With both present the two errors add. For the total mass, the first-order wander cancels only for zonal modes longer than the footprint, $k_{0n}R^* \ll 1$ (Eq. (6)).

In words: the middle of every column must sit close to the sheet plane, within $\varepsilon/2\pi$ of the shortest wavelength (0.8% of it for $\varepsilon = 5\%$). The column may be much thicker, up to $\sqrt{24\varepsilon}/2\pi$ of it (17%), because thickness is a second-order error and wander a first-order one.

Mass kernel: the upper bound

Feed the estimator 1 kg at a point $\mathbf{x}$. The data are its field $\mathbf{g}(\mathbf{x})$, and the answer is

$$w(\mathbf{x}) = \mathbf{h}^\top \mathbf{g}(\mathbf{x}), \quad \mathbf{h} = A^{+\top} \mathbf{f}_M. \quad (16)$$

The real data are a sum of such points, $\Delta \mathbf{d} = \int \mathbf{g}(\mathbf{x}) dm(\mathbf{x}) + \mathbf{n}$, and the true mass counts the points over the footprint, $\Delta M = \int 1_{R^*}(\rho) dm$. Since $\Delta \hat{M} = \mathbf{h}^\top \Delta \mathbf{d}$,

$$\boxed{\Delta \hat{M} - \Delta M = \int [w(\mathbf{x}) - 1_{R^*}(\rho)] dm(\mathbf{x}) + \mathbf{h}^\top \mathbf{n}.} \quad (17)$$

This is exact: no expansion, any mass distribution. *In words:* $w(\mathbf{x})$ is how many kg the estimator reports for each kg placed at $\mathbf{x}$. A perfect estimator has $w = 1$ under the footprint and $w = 0$ outside. Every bias is the gap $w - 1_{R^*}$, weighted by where the moved mass really is. Evaluating w at three heights of each column splits the bias exactly into the b_i of (1):

$$b_{\text{basis}} = \int (w_{z_0} - 1_{R^*}) dm, \quad b_{\text{wa}} = \int (w_{\text{mid}} - w_{z_0}) dm, \quad b_{\text{th}} = \int (w_{\text{col}} - w_{\text{mid}}) dm, \quad (18)$$

with w taken on the sheet plane (w_{z_0}), at the column middle (w_{mid}), and averaged over the column (w_{col}).

Closed form. Suppose the fit returned the exact Fourier–Bessel projection of the sheet, with no field-point fit and no SVD cutoff. A point mass m at radius ρ , in a column at $(\zeta, \Delta h)$, then gives the zonal coefficient $a_n = m J_0(k_n \rho) G_{k_n}/N_n$ (orthogonality for ρ , Eq. (8) for the height), with $k_n = k_{0n}$ and $N_n = \int_{R_\alpha} J_0^2 dA = \pi R_\alpha^2 J_1(j_{0n})^2$, $R_\alpha = \alpha R^*$. Wahr counts mode n as $W_n a_n$ with $W_n = \int_{R^*} J_0(k_n \rho) dA = 2\pi R^* J_1(k_n R^*)/k_n$; the modes with $m \geq 1$ integrate to zero over the footprint. Hence

$$\boxed{w_N(\rho, \zeta, \Delta h) = \sum_{n=1}^N c_n J_0(k_n \rho) e^{k_n \zeta} S(k_n \Delta h), \quad c_n = \frac{W_n}{N_n} = \frac{2R^* J_1(k_n R^*)}{k_n R_\alpha^2 J_1(j_{0n})^2}.} \quad (19)$$

The c_n are the Fourier–Bessel coefficients of the footprint 1_{R^*} . So on the sheet plane ($\zeta = \Delta h = 0$), w_N is the truncated series of a step, and its error is the Gibbs error:

$$\boxed{|w_N(\rho, 0, 0) - 1_{R^*}(\rho)| \lesssim \min \left\{ \frac{1}{2}, \frac{1}{\pi k_N |\rho - R^*|} \right\},} \quad (20)$$

exactly $\frac{1}{2}$ at the rim, a 9% overshoot on either side, then a 1/distance decay (checked numerically for TAG to within 20%, worst on the axis $\rho = 0$). *In words:* the three biases are the three factors of (19). Basis: with

N zonal modes the footprint edge is drawn as a ramp of width $\sim 1/k_N$, so mass near the rim counts half and mass a few metres away counts a few percent. This is not a thin-sheet effect. Wander and thickness: the per-mode gains $e^{k\zeta}$ and $S(k\Delta h)$ of the Result, now inside the sum. A gain on the short modes breaks the cancellation that makes the sum close to 1_{R^*} , so the wander is amplified, not averaged.

Each term in closed form. *Band limit (basis and outside).* The c_n make $w_N(\rho, 0, 0) = P_N 1_{R^*}$, the projection of the footprint on the first N zonal modes. P_N is symmetric, so with $\mu = \varrho \Delta h$ the mass per area,

$$b_{\text{band}} = \int (P_N 1_{R^*} - 1_{R^*}) \mu dA = \int_{R^*} (P_N \mu - \mu) dA, \quad (21)$$

the mass that smoothing the map to N modes moves across the rim. Near the rim ($k_N R^* \gg 1$) the ramp of (20) is the 1-D one, $w_N - 1_{R^*} \approx \pm(\frac{1}{2} - \text{Si}(k_N |\rho - R^*|)/\pi)$ (minus inside), whose integral is $1/(\pi k_N)$ on each side. Over the rim length $2\pi R^*$:

$$\boxed{b_{\text{in}} \approx -\frac{2R^*}{k_N} \bar{\mu}_{\text{in}}, \quad b_{\text{out}} \approx +\frac{2R^*}{k_N} \bar{\mu}_{\text{out}}, \quad \frac{b_{\text{band}}}{\Delta M} \approx -\frac{2}{\pi k_N R^*} \frac{\bar{\mu}_{\text{in}} - \bar{\mu}_{\text{out}}}{\bar{\mu}_*},} \quad (22)$$

with $\bar{\mu}_{\text{in}}, \bar{\mu}_{\text{out}}$ the mean mass per area in the rings of width $2/(\pi k_N)$ (the ramp's equivalent width) just inside and just outside the rim, and $\bar{\mu}_* = \Delta M / \pi R^{*2}$. A rigorous bound follows from Cauchy–Schwarz:

$$|b_{\text{band}}| \leq T_N \|\bar{\mu} - P_N \bar{\mu}\|_2, \quad T_N^2 = \pi R^{*2} - \sum_{n \leq N} \frac{W_n^2}{N_n} \approx \frac{2R^*}{k_N}, \quad (23)$$

with $\bar{\mu}(\rho)$ the ring average of μ , the only part 1_{R^*} sees. *In words:* N modes draw the rim as a ramp about $1/k_N$ wide, so they push mass across it. The error is the jump in mass per area across the rim, times the rim length, over πk_N . It vanishes if the rim sits where the map is flat ($\bar{\mu}_{\text{in}} = \bar{\mu}_{\text{out}}$), and it falls as $1/(k_N R^*)$.

Fit (aliasing and cutoff). With the SVD cutoff, $A^+ A = V V^T$ (V : the kept right singular vectors). Split the field of 1 kg as $\mathbf{g}(\mathbf{x}) = A\boldsymbol{\phi}(\mathbf{x}) + \mathbf{r}(\mathbf{x})$, with $\boldsymbol{\phi}$ the coefficients of the exact projection ($\mathbf{f}_M^T \boldsymbol{\phi} = w_N$) and $\mathbf{r}$ what no mode reproduces at the field points. Then

$$w - w_N = -\mathbf{f}_M^T (I - V V^T) \boldsymbol{\phi}(\mathbf{x}) + \mathbf{h}^T \mathbf{r}(\mathbf{x}). \quad (24)$$

The first term is the part of the Wahr weights in the discarded directions (cutoff); the second is out-of-basis field that the fit reads as CH signal (aliasing). Both need the truth's own coefficients and field: this term has no closed form, and only the kernel measures it.

Mesh vs point masses. Each entry of $\mathbf{g}$ is a potential or a gradient of $1/|\mathbf{p} - \mathbf{x}|$, so w is harmonic in the source position, $\nabla^2 w = 0$. Averaging w over a cell $a \times a \times \Delta h$ changes it only at second order, by $\frac{a^2}{24} \nabla_h^2 w + \frac{\Delta h^2}{24} \partial_z^2 w$, with $\nabla_h^2 w = -\partial_z^2 w$:

$$\boxed{\bar{w}_{\text{cell}} \approx w + \frac{\Delta h^2 - a^2}{24} \partial_z^2 w, \quad b_{\text{th}} \approx \frac{1}{24} \int \Delta h^2 \partial_z^2 w dm, \quad b_{\text{cell}} \approx -\frac{a^2}{24} \int \partial_z^2 w dm.} \quad (25)$$

Per mode $\partial_z^2 = k^2$: the column gives the thickness gain $S(k\Delta h) \approx 1 + (k\Delta h)^2/24$ of Eq. (10), and the cell gives $1 - (ka)^2/24$. So no mode is off by more than $(k_{\text{max}} a)^2/24$. *In words:* a cell and a point mass differ only by the cell's quadrupole. The thickness is its vertical half, the grid spacing its horizontal half with the opposite sign; a cube has none.

The bound. Since $|\int e dm| \leq \int |e| |dm|$, (17) gives

$$\boxed{|\Delta \hat{M} - \Delta M| \leq \int |w - 1_{R^*}| |dm| + 2\sigma \leq \max |w - 1_{R^*}| M_{\text{moved}} + 2\sigma, \quad M_{\text{moved}} = \int |dm|.} \quad (26)$$

It holds for any density in the moved volume. The price is M_{moved} : excavated and deposited mass are added, not allowed to cancel. With the pipeline's uniform density, $dm = \varrho \Delta h dA$, the b_i of (18) are signed and known. If the density departs from ϱ by at most a fraction δ_ϱ , each dm moves by at most $\delta_\varrho |dm|$, so

$$|\Delta \hat{M} - \Delta M| \leq \sigma_{\text{sys}} + \kappa \delta_\varrho |\Delta M| + 2\sigma, \quad \sigma_{\text{sys}} = \sum_i |b_i|, \quad \kappa = \frac{\int |w - 1_{R^*}| \varrho |\Delta h| dA}{|\Delta M|}. \quad (27)$$

In words: κ says how much each percent of density variation can move ΔM . $\delta_\varrho = 0$ is the uniform-density budget (2); $\delta_\varrho = 1$ allows any density between 0 and 2ϱ . Every term is the estimator's own kernel integrated against the DTMs: no simulation, and no knowledge of the true mass.

TAG: error budget

TAG basis $(\alpha; M, N) = (6; 8, 22)$, $R^* = 8$ m: $k_{\text{max}} = 1.65 \text{ m}^{-1}$, $\lambda_{\text{min}} = 3.8$ m. For $\varepsilon = 5\%$, Eq. (15) asks $|\Delta h| \leq 0.67$ m and $|\zeta| \leq 3$ cm. A typical TAG column (median over the moved volume) has $|\Delta h| = 0.36$ m and $|\zeta| = 0.36$ m: the thickness passes, the wander fails by $\times 12$. So expect a negligible thickness bias and a wander bias at the percent level.

Each b_i is Eq. (18) with the pipeline's own kernel $w = \mathbf{h}^T \mathbf{g}$ (same 10 000 field points, same SVD cutoff: 64 of 352 singular values kept), evaluated at every 0.3-m DTM column with $dm = \varrho \Delta h dA$. Running the pipeline's fit on the field of the same point masses gives the same numbers; the mesh row is the gap between these point masses and the polyhedral field of the real run.

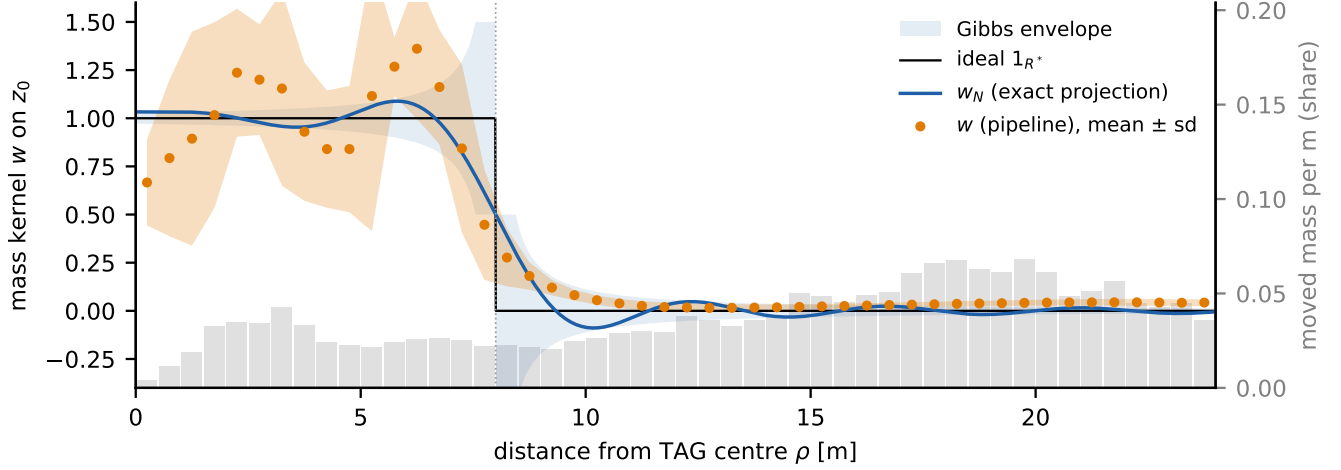
source of error	in σ ?	b_i (uniform ϱ)	bound, any ϱ
noise	yes	$\pm 0.31\%$ ($1\sigma = 87$ kg)	
thickness, Eq. (5)	no	-0.13%	2%
wander, Eq. (4)	no	$+2.40\%$	73%
basis: band limit + SVD cutoff	no	$+2.77\%$	54%
material outside R^*	no	$+0.08\%$	25%
mesh vs point masses	no	$+0.19\%$	—
sum		$+5.31\%$	$\kappa = 104\%$
actual error $\Delta \hat{M} - \Delta M$		$+5.30\%$ (17σ)	

The last column is $\int |\Delta w| |dm| / |\Delta M|$ for each row; κ is the same integral for the whole $w - 1_{R^*}$, smaller than the sum of the rows. With uniform density, Eq. (27) gives $\sigma_{\text{sys}} = 5.6\%$ and

$$\Delta M = -2.98 \times 10^4 \text{ kg} \pm 87 \text{ kg (stat)} \pm 1.7 \times 10^3 \text{ kg (sys)}, \quad (28)$$

which contains the true -2.83×10^4 kg: $5.3\% \leq 5.6\% + 2(0.31\%)$. A density that varies by $\pm \delta_\varrho$ between columns adds up to $1.04 \delta_\varrho$: $\pm 10\%$ in density allows 10% more in ΔM .

Why $\kappa \approx 1$. TAG moved far more mass than it removed: $M_{\text{moved}} = 7.3 |\Delta M|$ over the patch ($4.1 |\Delta M|$ excavated, $3.2 |\Delta M|$ deposited). The kernel error averaged over the moved mass is 0.14, and $0.14 \times 7.3 = 1.04$. Uniform density makes excavation and deposition cancel to 5%; without it nothing forces them to.



Mass kernel on the sheet plane versus distance from the TAG centre. Black: a perfect estimator. Blue: the closed form (19) with the Gibbs envelope (20). Orange: the pipeline's kernel, mean $\pm$ s.d. over each 0.5-m ring of DTM columns. Grey: where the moved mass is.

The basis error. The closed form (19) gives the band limit alone: -1.94% from mass inside R^* and $+1.07\%$ from outside, both from the Gibbs ramp at the rim. The pipeline gives $+2.77\%$ and $+0.08\%$. The difference is the fit. The 352 modes cannot match the sheet's field exactly at the field points (5% of it is left in the residual even without noise), so the fit bends the coefficients, the zonal ones included, to make up the rest (aliasing). In the figure this is the ± 0.3 – 0.5 scatter of w around w_N inside R^* . The SVD cutoff decides how much gets through. Without noise:

SVD cutoff (values kept of 352)	8.3×10^{-3} (64)	10^{-4} (119)	10^{-6} (164)	10^{-8} (205)
b_{basis}	$+2.77\%$	$+0.66\%$	-18.5%	$+18.0\%$
$\sum b_i$	$+5.1\%$	$+1.8\%$	-21.6%	-34.5%
κ	1.04	1.87	2.67	5.37

A looser cutoff first lowers the bias, then lets aliasing through, and κ grows throughout. With the exact projection and all 22 zonal modes (w_N , no cutoff), the wander alone is $+211\%$ and $\kappa = 12$: the cutoff is what removes the large $e^{k_n \zeta}$ of the short modes. The noise-only σ cannot choose the cutoff; the kernel can.

Formulas against the kernel.

term	formula, TAG inputs	formula	exact (kernel)
band limit, inside R^*	Eq. (22), $\bar{\mu}_{\text{in}} = 0.39 \bar{\mu}_*$	-2.18%	-1.94%
band limit, outside R^*	Eq. (22), $\bar{\mu}_{\text{out}} = 0.22 \bar{\mu}_*$	$+1.25\%$	$+1.07\%$
band limit, total	bound (23), $T_N = 3.34$ m	$\leq 6.3\%$	-0.87%
fit: aliasing + cutoff	none	–	$+3.72\%$
thickness	Eq. (25)	-0.12%	-0.13%
cell vs point ($a = 0.30$ m)	Eq. (25)	-0.03%	

The rim formula is within 20% of the exact band-limit terms. At TAG's rim the mass per area is $0.39 \bar{\mu}_*$ just inside and $0.22 \bar{\mu}_*$ just outside; the jump, $0.17 \bar{\mu}_*$, times $2/(\pi k_N R^*) = 0.056$ gives -0.9% . The rigorous bound (23) is 6.3% ($\bar{\mu}$ in rings one grid cell wide; 18% of its norm lies above the band). The fit, pipeline minus exact projection ($+2.85\% - (-0.87\%)$), has no formula and is the largest part of the basis term. The cell term is -0.03% (at most $(k_{\text{max}} a)^2/24 = 1\%$ per mode); the rest of the mesh row, $+0.2\%$, is where the 0.3-m DTM height maps and the meshes differ, a mass-map error covered by the caveat below.

Caveats. The b_i assume uniform ρ and exact DTMs. Any error δm in the mass map adds at most $\int |w - 1_{R^*}| |\delta m|$ (for a DTM error δh , put $|\delta h|$ for $|\Delta h|$ in κ). 0.9% of the field points are within 0.5 m of the sheet, where w is grainy. 0.14% of the field points lie below z_0 (down to -0.21 m). One random draw of field points; w depends on where they are.

So: the noise is 0.3% and the systematics 5.6%; that is why the error is 17σ and not 1σ . Half of the systematics is the thin sheet, almost all of it the wander, as the conditions predicted; the thickness is -0.1% . The other half is the basis, mostly the fit (aliasing and cutoff): the band limit alone is -0.9% . The analytical upper bound is

$$|\Delta\hat{M} - \Delta M| \leq 5.6\% + 1.04\delta_\varrho + 0.6\% \quad \text{of } |\Delta M|, \quad (29)$$

so the TAG mass is as good as the uniform-density assumption. Every b_i comes from the DTMs, so it can also be removed instead of quoted: the terrain refinement, which puts the sheet on the real surface, takes ΔM from $+5.3\%$ to $+1.1\%$.

[D] G. Fereoli, *Dissertation, Part 2: Cylindrical Harmonics*, <https://github.com/giovannifereoli/Dissertation---CH-Pt.2>. Numbers: `notes/ch_thin_sheet_tag_check.py` (conditions), `notes/ch_thin_sheet_budget.py` (budget, kernel, bounds, figure).